

Answer Key

Question 1

Sortino ratio = (Average return – MAR) / Downside deviation

$$\text{Sortino ratio} = (14\% - 6\%) / 8\% = 8 / 8 = 1.00$$

Question 2

1. Downside deviation

Only returns falling below the 3% MAR contribute. For each quarter, take the minimum of 0 and (return – MAR):

Quarter	Return	Return – MAR	Contributes?
Q1	7%	+4%	No (0)
Q2	-4%	-7%	Yes (-7%)
Q3	11%	+8%	No (0)
Q4	2%	-1%	Yes (-1%)
Q5	-8%	-11%	Yes (-11%)
Q6	14%	+11%	No (0)

$$\text{Downside deviation} = \sqrt{[\text{mean}(0, 0.0049, 0, 0.0001, 0.0121, 0)]} = \sqrt{0.00285} \approx 5.34\%$$

2. Average return and excess return

$$\text{Average return} = (7\% - 4\% + 11\% + 2\% - 8\% + 14\%) / 6 = 22\% / 6 \approx 3.67\%$$

$$\text{Excess return} = 3.67\% - 3\% (\text{MAR}) \approx 0.67\%$$

3. Sortino ratio

$$\text{Sortino ratio} = 0.67\% / 5.34\% \approx 0.12$$

4. Interpretation

A Sortino ratio of 0.12 is very weak. The portfolio's average return (3.67%) barely clears its own minimum acceptable return (3%), leaving almost no cushion of excess return.

Meanwhile, two of the six quarters (Q2 and especially Q5) produced sizeable shortfalls below the MAR, driving downside deviation up to over 5%. The result is a portfolio that took on meaningful downside risk relative to the modest excess return it delivered. A risk-averse investor would likely see this as inadequate compensation for the risk of falling short, even though the portfolio's raw average return looks acceptable on the surface.